My First Maze: Make a Game You Can Play on a Phone

How to Use This Handout

You have about 90 minutes. Do each part on your screen, then tick the box. Stuck? Ask a neighbour first, then your teacher. At the end you will make a game that you can play in a web browser, and even on a phone!

Part 1: Open Your Maze (5 min)

- [ ] Click **File > Open Project** and open the folder **My Maze**.
- [ ] Look at the list on the left. You have a **spr_wall** (picture), an **obj_wall** (the wall) and a room called **room0**.
- [ ] Press the green triangle (or **F5**) to test. You see an empty box. Close the game window.

> Press **Save** often. The disk button saves your game.

Part 2: Draw Your Hero and Your Goal (15 min)

- [ ] Click **Assets > Create Sprite**. Name it spr_hero, then OK.
- [ ] Draw your hero! Fill the square. Click **Save**.
- [ ] Do it again for the goal. Name it spr_goal. Make it look different from the hero and the walls.

> Names use letters, numbers and _ only. No spaces!

Part 3: Make the Hero Move (20 min)

- [ ] Click **Assets > Create Object**. Name it obj_hero.
- [ ] In **Sprite**, choose spr_hero.
- [ ] In the **Blockly** area, build these blocks. Drag them from the left.

When this happens	Do this
Keyboard: Left Arrow (held)	**Set horizontal speed to** -4
Keyboard: Right Arrow (held)	**Set horizontal speed to** 4
Keyboard: Up Arrow (held)	**Set vertical speed to** -4
Keyboard: Down Arrow (held)	**Set vertical speed to** 4
Keyboard: No key	**Stop movement**
When colliding with obj_wall	**Stop movement**

- [ ] Click **Save**.

> Right-click a block to **Duplicate** it. It is faster than dragging a new one every time.

> A minus sign makes the hero go left or up. A plus number goes right or down.

Part 4: Make the Goal (10 min)

- [ ] Create an object named obj_goal. Give it spr_goal.
- [ ] Build these blocks:

When this happens	Do this
When colliding with obj_hero	**Show message** "You win!"
(the same event)	**Restart room**

Put the two blocks one under the other, under the same event.

Part 5: Build Your Maze (20 min)

- [ ] Open **room0** (double-click it in the list).
- [ ] Click **obj_wall**. Hold the mouse button down and **drag** to paint a wall.
- [ ] Click **obj_hero** and click once inside the top-left of the room.
- [ ] Click **obj_goal** and click once far away from your hero.
- [ ] Press **F5** and play! Can you reach the goal?

> Make **every path two squares wide**. Your hero is one square wide and does not always stop in the middle of a path. In a path of only one square the hero can get stuck.

> You should see your hero, your goal and your walls. The hero stops at a wall. When you touch the goal, "You win!" appears.

Part 6: Share Your Game (10 min)

- [ ] Click **File > Export as HTML5**.
- [ ] Choose a folder. Click **Yes** to open the game in your browser.
- [ ] Look in the folder: you have **one file** ending in .html. That is your game!
- [ ] Play it. Then show it to a friend.

> On a phone or a tablet, arrow buttons appear on the screen. Hold a button to move.

Stuck?

What happens	Try this
The hero does not move	Is the hero in the room? Is there a block under **Keyboard**?
The hero never stops	Add **Keyboard: No key** with **Stop movement**
The hero goes through walls	Add **When colliding with** obj_wall with **Stop movement**
The hero is stuck in a path	Make the path two squares wide
Nothing happens at the goal	The goal needs **When colliding with** obj_hero
I deleted something by mistake	**Tools > Restore Deleted Assets**

Words to Know

- **Sprite**: a picture
- **Object**: something in your game that can do things
- **Room**: a level
- **Event**: "when something happens"
- **Collision**: when two things touch
- **Export**: make a copy of your game that works on its own

Challenges

- **Try this (5 min):** change the message to something funny.
- **Push further:** make a second goal, or draw a better wall sprite.
- **Invent:** swap computers with a friend and try to finish their maze.

Think About It

1. Why must the hero stop when it touches a wall?
2. What does the goal do when the hero touches it?
3. Why do we make the paths two squares wide?

My Notes